Tottie

Self-Curating Agent Memory over MCP

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A year of conversations. It still doesn't know your name.

The real problem

Agents do not have a reasoning problem. They have a **state persistence** problem.

- Every agent wakes up stateless: no history, no context, no past.
- A bigger context window is not memory — it is an expensive scratchpad.
- Longer context $\Rightarrow$ linearly higher cost, super-linear latency, *lower* accuracy.

"All that intelligence, and it fails the baseline of a mind: continuity."

Everyone searches at read time. We think at write time.

Standard RAG

Intelligence at **read** time: the agent searches when it needs something. Memory is a pile of notes the agent must juggle.

Tottie

Intelligence at **write** time: memory curates itself as you live. The agent does not manage memory — it just remembers.

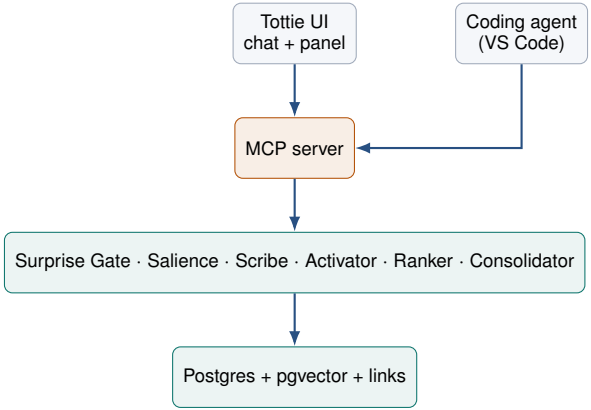
This is not a “basic RAG application.” The value is the **ingestion-time decision**, the deliverable is an **MCP server**, and retrieval is **associative**, not just similar.

What Tottie is

A self-curating memory architecture that runs the memory lifecycle **in parallel** with the agent, and ships as one **MCP server** any agent can mount.

- Positioned against 2026's **learned consolidation** field (mem0, Zep-Graphiti, A-Mem).
- Differentiator 1: write-time salience + **reconciliation** is the product, not a bolt-on.
- Differentiator 2: **transport-agnostic** — the same brain in a chat, an IDE, anything.
- Differentiator 3: retrieval is **cue-driven associative recall**, not one similarity search.

We did not invent the design. We copied it.



- **Surprise Gate** — predictive coding: encode only what we did not expect.
- **Activator** — spreading activation, priming, decay.
- **Ranker** — we do not forget; we re-rank.
- **Consolidator** — sleep: generalise episodes into lessons.

The memory vault has tiers

Tier	What it holds
Identity	Persona + permanent constraints. Pinned raw, zero search.
Episodic	Chronological decisions, state changes, history.
Knowledge	Ingested docs, files, references (semantic).
Procedural (“Lessons”)	Mistake $\rightarrow$ solution pairs. Autosave.
Prospective	Intentions with an event or time cue.

Why tiers beat a flat vector store

Identity needs no search at all; episodic needs time; knowledge needs semantics; lessons need an error signature. One store, five retrieval policies.

The part everyone leaves out: what reaches the model?

Dual-path recall: a card handed over automatically, plus a probe the agent can run.

The Context Card (injected every turn)

```
[MEMORY CARD - 3 hits]
▷ IDENTITY (pinned) user=Frans - girlfriend birthday = Nov 7
▷ EPISODIC 2026-10-29 decided:  pgvector, not Qdrant #tottie #decision
▷ KNOWLEDGE (telecom-arch.pdf p.7) Server C behind A/B #telecom #infra
→ why:  cue=COMMAND → seed #tottie → spread 1 hop
```

- **Ambient** — the card is always there; the agent "just knows."
- **Deliberate** — the card carries pointers; `memory.probe(tag)` opens the room.

It changes its mind, correctly

Contradiction at write time, not read time

"I'm vegetarian now" $\rightarrow$ the Scribe **supersedes** the old fact: old row `valid_to` $=$ *now*, new row `valid_from` $=$ *now*.

- Retrieval filters `valid_to IS NULL`, so a stale fact can never surface.
- A retrieved memory becomes *labile* — contradict it now and it is **rewritten** (reconsolidation) — but **nothing is lost**: the old version is kept and versioned.
- Exactly LongMemEval's *knowledge update* and MemoryAgentBench's *conflict resolution*.

It does not search. It associates.

“Implement feature A” is not a question

It is a **COMMAND cue**. Its policy is: *assemble what a competent person needs to act* — project bindings, conventions, past decisions, lessons, constraints.

- From the cue’s nodes, **activation spreads** along memory links and **decays** with distance.
- Then **pattern completion** reconstructs the whole coherent context.
- Three legs, fused: **semantic + lexical + associative**.
- This is why a memory from fifteen years ago surfaces the moment someone mentions a band.

RAG is one leg. This is the whole animal.

The constraint we refused

Simulate the brain's solutions. Never simulate its constraints.

The brain forgets because it is **bounded** — finite neurons, metabolic cost, lossy reconstructive storage. Those are *constraints, not features*. We are not bounded: storage is nearly free and search is exhaustive.

- We **copy** the solutions: spreading activation, consolidation, pattern completion, surprise gating, reconsolidation, salience.
- We **refuse** the constraints: passive decay, lossy recall, retrieval blocking, a 7-item scratchpad.

Copying Ebbinghaus decay would be copying a limitation to solve a solution problem.

But what about forgetting? It is ranking, not deletion.

Availability $\neq$ accessibility (Tulving & Pearlstone)

Available = stored. **Accessible** = retrievable *right now*, and that depends on whether the cue matches the encoding. Tip-of-the-tongue is a failure of accessibility.

Human

Memory fades, the cue fails, and it is gone. The feeling of knowing is a *bug*.

Tottie

Everything stays *available*; only the cued subset is *accessible*. We keep the mechanism and drop the bug — **the cue always works**.

So we renamed it: the Decayer is a **Ranker + Compressor**, not a shredder.

It only remembers the surprising — and it never shreds.

Surprise gating (predictive coding)

The brain encodes the *prediction error*, not the input. Tottie computes a surprise score and refuses to store the routine: *only the unexpected is written*.

Ranking (“forgetting”)

```
utility = exp(-ln2*age/half_life) * (1 + log(1+access_count))
```

- Unused memories leave the *active set*; used ones strengthen (the testing effect).
- Nothing is destroyed. A deprioritised memory returns the instant the right cue arrives.
- *Forgetting is the feature. Deletion is not.*

Then it sleeps — and generalises.

The Consolidator (“Sleep”)

Replays recent episodic memories, extracts the recurring structure into **semantic** knowledge, and compresses what is never used. *12 episodic → 3 semantic; 4 ranked out.*

- The neurological analogue: hippocampal replay during sleep builds neocortical knowledge.
- No other memory tool shows *generalisation* on camera.
- On demand, so it is deterministic and demoable.

It remembers the future, and it knows what it doesn't know.

Prospective memory

Intentions with an **event** or **time** cue fire proactively — "remind me to ask about the deploy on Friday."

Metacognition

A **confidence** score from the retrieval margin. Low confidence $\rightarrow$ verify, ask, or abstain. The agent knows when it does not know.

A memory store returns facts. A mind returns the right facts, at the right time, with a sense of how sure it is.

One brain. Three bodies.

Tottie

Chat + reacting sprite.
Remembers your *life*.

Coding agent

Same MCP server in VS Code.
Knows your *stack*; autosaves fixes.

Memory Panel

The live X-ray. Watch it think.

The money shot: add a fact in the chat, then ask your coding agent — it already knows.

The Memory Panel: the invisible, made visible

Real time, four blocks:

1. **Learning now** — Observer verdict (KEEP/DISCARD, salience, **surprise**).
2. **Memories formed** — `NEW` / `UPDATE` / `SUPERSEDE` / `SKIP`, with the text.
3. **Where it inserted them** — tier chip + tag/room chips + a growing room map.
4. **Why this surfaced** — `cue=COMMAND` $\rightarrow$ `seed` $\rightarrow$ `spread 1 hop`, plus the exact card received.

Why this is the hero

Judges should not have to take “it remembers” on faith. They watch the decision happen.

We do not ask you to trust it. We measure it.

- A ∼**40-item** suite across LongMemEval's categories (knowledge update, temporal, preference) plus contradiction resolution and abstention.
- Every item runs **memory ON vs OFF** — the ablation that proves memory caused the answer.
- Metrics: answer accuracy, memory-hit@k, contradiction-correctness, tokens/query, p50/p95 latency.

The headline

91% lower p95 latency and **90% fewer tokens** than full-context, for a small accuracy trade — the write-time curation argument, made falsifiable.

What was built during the Hackathon (Oct 27 – Nov 4)

Component	
`core/`	Engine: Surprise Gate, Salience, Scribe, Activator, Ranker, Consolidator, Reconciler, Prospective Queue
`mcp/`	MCP server: 8 tools, 2 resources, the context-card prompt
`ui/`	Tottie chat, sprite state machine, live Memory Panel, why-surfaced, Sleep, ablation
`eval/`	ON/OFF ablation harness + consolidation check
`docs/`	Cognition model + architecture + flows + this deck (LaTeX $\rightarrow$ PDF)

Every component is committed on a dated branch; the ledger is in `docs/business/hackathon_compliance`.

So what did I actually build it for?

"Everything I was mad at my girlfriend for —
and the one thing that matters: her birthday."

- **Autosave** — an agent makes a mistake once, and never again.
- **A personal AI** that actually knows you, not just your last 20 messages.
- **The company brain** — remember that Server C sits behind A and B before designing it.

Tottie

It just remembers.

github.com/frans-nekongo/Tottie